

The Way FreeBSD Jails Work



FreeBSD-based computer systems can be partitioned into several independent mini-systems called jails by using operating system level virtualisation, which is an implementation of FreeBSD. FreeBSD jails compartmentalise the system – its files and its resources—in such a way that only the right person has access to the right compartments (jails).

Currently, container virtualisation is very popular. It is used for PaaS (Platform as a Service) and IaaS (Infrastructure as a Service), although usage for the former is more common.

PaaS makes application deployment seamless and very easy for developers, who do not have to deal with the intricacies of server set-up and management, which can often be overwhelming.

FreeBSD is an excellent operating system for servers because its network performance (if tuned properly) trumps all other operating systems, and it has the Z File System. ZFS makes the life of a systems administrator extremely easy. There is ZFS on Linux as well, but due to licensing issues, Linux will never get native ZFS. The last time I tried, it was quite buggy and for some reason, the ZFSOnLinux developers had declared the version I was using as stable!

What is container virtualisation?

To quote Wikipedia, “Operating system–level virtualisation is a server virtualisation method where the kernel of an operating system allows for multiple isolated user space instances, instead of just one. Such instances (often called containers, virtualisation engines (VE), virtual private servers (VPS), or jails) may look and feel like a real server from the point of view of its owners and users.”

If you have used something like KVM, VirtualBox, etc, you might recall that you had to install a complete operating system

from the installation disk, just as you would do for a physical machine. This mode is known as full virtualisation – the hardware, like the disk, CPU, etc, is emulated by the hypervisor (the program that runs on the host system) for the guest operating system.

The advantage of this mode is that two guests and the host are completely isolated from each other and it is even possible to run multiple operating systems on a single host machine, i.e., you can have a physical Linux server and run Linux, FreeBSD and Windows as guest operating systems over it.

The disadvantage of full virtualisation is that, it is not always beneficial. There are performance issues since there is emulation of a complete machine’s resources, and if you want to secure a single service – say you want to run an http server isolated from everything else, or a database server isolated from everything else – it is a pure overkill. You can’t use the full potential of your hardware in such cases.

For most use cases like these, container virtualisation is more than sufficient. There is no doubt that container virtualisation is the future of cloud computing with technologies like FreeBSD Jails, Docker (LXC or Linux Containers), etc.

Container virtualisation is implemented on UNIX-like systems using the chroot system call, by which it is possible to switch to a specific path on the file system, and have your shell and further programs consider the root as the path you are currently in.

There is a single kernel which runs on the host machine. It is not possible to run different operating systems (i.e.,